

SpaceLoop Engineering // Implementation Dossier

Comprehensive Daily Technical Log & Feature Deliverables • Date: September 28, 2026 • Sprint: Hack2Ignite

Platform: SpaceLoop P2P Micro-Leasing	Stack: React 18, Vite, TS, Tailwind, Flask, SQLite	Target Hosting: Vercel (Edge UI) + Render (API)
Active Remotes: origin, render, vercel	AI Routing: Groq → Gemini → Rule Fallback	Compliance: Sec 52 Easements Act, Zero-Hallucination

1. Executive Summary of Implementations

During this engineering cycle, the team executed high-impact features spanning natural language processing (NLP), retrieval-augmented generation (RAG), user experience refinements, and production edge synchronization. All changes strictly respected the core P2P workflow (Host → Listing → Search/Match → Booking → Access → Checkout/Escrow) and architectural rules (multi-tier LLM routing, deterministic core logic, zero false pretenses, and no framework migration).

Feature / Module	Scope & Subsystem	Key Technical Deliverables	Status
NLP Part 2: Semantic Search	Backend AI & Search Layer	Neural query parsing, concept expansion (WiFi, Quiet, AC), composite match scoring, explainable match justifications.	Completed & Verified
NLP Part 3: Loop Bot RAG	AI Assistant / Loop Bot	Grounded retrieval over real listing inventory, hourly pricing, amenities, Section 52 legal context; anti-hallucination guardrails.	Completed & Verified
NLP Part 4: Listing Assistance	Host Creation & Extraction	Unstructured text-to-listing drafting, multi-tier Groq/Gemini routing, word-boundary grounded amenity filtering, host review UI in ListSpacePage.	Completed & Verified
Portal Navigation Cleanup	Header & Drawer (Seeker/Host)	Moved secondary actions (My Bookings, How It Works, Yield Calc, KYC, List Space) out of navbar into draggable NavigationDrawer.	Completed & Verified
Portal Indicator Redesign	Header UI / Brand Segment	Replaced oversized HOST PORTAL badge with sleek, hardware-inspired illuminated LED pill ('• Host Mode' / '• Seeker Mode').	Completed & Verified
Team Architecture Page Fixes	Editorial Showcase Page	Clean bottom section boundaries, hairline architectural guides, optical cursor tracking, and member profile photo alignments.	Completed & Verified
Production Sync & Vercel Deploy	DevOps & Edge Delivery	Compiled modern Vite bundles and synced directly to public/distribution assets for instantaneous Vercel edge deployment.	Deployed Live

2. SpaceLoop NLP Part 2: Semantic Search & Query Understanding

Problem Statement: Traditional search required exact keyword matches. When seekers searched for conceptual descriptions (e.g. *'quiet place to work with fast internet'*), relevant listings offering high-speed Wi-Fi, air-conditioned cabins, or acoustic quiet zones were missed if exact tokens were absent.

Architectural Solution & Implementation:

- Neural Concept Expansion:** Gemini and deterministic fallback heuristic extract underlying seeker intent, mapping queries to conceptual tags (e.g. 'Wi-Fi' → 'Fiber', 'High-Speed Internet'; 'Quiet' → 'Focus Pod', 'Study Room').
- Composite Proximity & Attribute Matcher:** Incorporates Haversine geographical proximity, budget constraints, capacity alignment, and amenity overlap.
- Explainable Output:** Returns a transparent match_score (0–100%) and an explainable *'Why this matches'* justification for user confidence.

- **Zero Parallel Systems (Rule 8):** Fully integrated into existing SQLAlchemy listings database without spinning up external search clusters or vector silos.

3. SpaceLoop NLP Part 3: Retrieval-Augmented Generation (RAG)

Objective: Empower Loop Bot to answer detailed, SpaceLoop-specific queries using real, authoritative application data rather than hallucinating generic responses from LLM pre-training weights.

Retrieval Pipeline

RAG Pipeline Flow:

User Query → Query Understanding & Intent Classification → Grounded SpaceLoop In-Memory Retrieval → Context Assembly & System Guardrails → Multi-Tier LLM → Fact-Grounded Response → Loop Bot UI

Key RAG Guardrails:

- **Inventory Grounding:** Fetches real database listings with live hourly rates, neighborhood data, and capacity.
- **Legal & Trust Grounding:** Automatically retrieves Section 52 of the Indian Easements Act parameters, ₹100 refundable UPI micro-escrow rules, and Discom CA meter validation facts.
- **Anti-Hallucination Barrier:** The bot refuses to invent pricing, promise unavailable bookings, or fabricate unverified contact details.

4. SpaceLoop NLP Part 4: Listing Assistance & Zero-Hallucination Guardrails

Objective: Allow property hosts to enter natural language descriptions (e.g. *'2 bedroom workspace in Pune for 4 people with fast WiFi and AC'*) and automatically generate a structured, type-safe listing draft mapped to the Space database model.

Zero-Hallucination Implementation Highlights:

- **Elimination of Synthetic Guesses:** Removed legacy heuristics that fabricated square footage ($550 + (\text{bedrooms} - 1) * 350$) and capacity ($\text{bedrooms} * 2$). Unmentioned fields remain strictly null / None.
- **Word-Boundary Regex Grounding:** In `_parse_llm_json`, implemented strict regex word boundary checks (`(\bac\b, \bwifi\b)`). This fixed a critical edge case where the letters 'ac' inside the word 'workspace' falsely triggered Air Conditioning detection.
- **Human Confirmation Flow (Rule 4):** The AI never auto-publishes. Extracted attributes populate the host form in `ListSpacePage.tsx` with missing fields flagged as **Unspecified** in amber. The host reviews and clicks *'Publish Active Space Listing'* to finalize.
- **1-Click Test Prompts:** Built 1-click test chips directly into the UI for rapid testing of sparse vs. complete prompts in English and Hindi.

5. Multi-Tier LLM Architecture & Resilience Strategy

Following **Rule 5 (AI Should Assist the Product, Not Control It)**, all AI features utilize a deterministic, multi-tier failover orchestrator guaranteeing 100% platform availability even during cloud API outages.

Tier Level	Provider / Engine	Model & Configuration	Role & Failover Trigger
Tier 1 (Primary)	Groq Cloud	llama-3.3-70b-versatile JSON Mode, temp=0.1	Ultra-low latency extraction and conversational grounding. First priority.
Tier 2 (Fallback)	Google Gemini	gemini-2.5-flash Structured prompt, temp=0.1	Triggers automatically on Groq HTTP 429 rate limit or network timeout.
Tier 3 (Core)	Deterministic Rules	Regex parsing & Indic property dictionaries	Guarantees platform operation when both cloud APIs are offline or keyless.

6. Navigation System Overhaul & Drawer Engine

To prevent cognitive overload on desktop and eliminate horizontal crowding, the navigation system underwent a complete architectural refactoring across both the Seeker and Host portals.

Portal	Retained in Main Navbar	Relocated to Hamburger Drawer
Seeker Portal	- SpaceLoop Logo - Compact Seeker Mode Indicator - Explore Spaces (/explore) - Architecture Team (/architecture) - Switch to Host Portal - Seeker Sign In / Student SSO	- My Bookings (/dashboard) - How It Works (/how-it-works) - Trust & Safety Console (/admin/trust-safety) - Language Selector dropdown
Host Portal	- SpaceLoop Logo - Compact Host Mode Indicator - Host Dashboard (/host/dashboard) - Architecture Team (/architecture) - Switch to Seeker Portal - Language Selector - Theme Toggle (Ocean Breeze / Midnight Neon) - Host Sign In / Register Space	- List a Space (/list-space) - Yield Calculator (/calculator) - Discovery & KYC (/verify) - Account profile controls - Dedicated Seeker switcher CTA

Physical Navigation Drawer Features (NavigationDrawer.tsx):

- 60fps Physics & Edge Swipe:** Integrated pointer capture allowing users to drag open the menu from screen edges with spring settling animation.
- Dual-Theme Compliant:** Automatically syncs with Ocean Breeze (Light) and Midnight Neon (Dark) design palettes.
- Zero Duplication Guarantee:** Items in the drawer do not repeat on the main navbar.

7. Portal Indicator UI Refinement

The Issue: The previous header placed a large, standalone rectangular badge reading HOST PORTAL separated by a harsh vertical divider, visually competing with Host Dashboard and primary CTA buttons.

The Redesign: Replaced the oversized badge with a sleek, minimalist hardware-inspired pill indicator: [• Host Mode] / [• Seeker Mode].

- Proximity:** Positioned directly beside the SpaceLoop brand text without arbitrary border dividers.
- Illuminated LED Indicator:** Steady glowing hardware LED (Amber for Host, Indigo for Seeker) without distracting double-ping CSS keyframes.
- Geometry & Contrast:** Fully rounded pill (rounded-full px-2.5 py-0.5 text-[11px]) on translucent backdrop with subtle neon ambient glow.

8. Architecture & Engineering Team Section Layout Polish

Addressed the visual collision below the 4th architecture card where subsequent content clashed with card drop-shadows:

- **Clean Section Boundary:** Added structural bottom border and architectural hairline guide divider (ag-guide-line-h).
- **Responsive Breathing Room:** Expanded bottom clearance (pb-20 sm:pb-28 lg:pb-32) accommodating 3D card tilt and hover elevations.
- **Verified Team Members:** Preserved all 4 verified engineer profiles: Arya Maurya, Kanishk Singh, Zara Quadri, and Indrayani Mazumder.

9. Production Deployment & Edge Synchronization

Vercel serves SpaceLoop directly via edge-cached static distribution files in public/. A key DevOps task was compiling the newest production build and synchronizing public assets.

Target Environment	Deployment URL	Asset Bundle Hash	Sync Verification
Vercel Production	https://spaceloop.vercel.app	JS: index-DMUZ9VF1.js CSS: index-BAGGljd5.css	Verified via Live HTTP
Render Backend	https://spaceloop.onrender.com	Python Flask API + SQLite DB	Automated GitHub Sync
GitHub Core Repo	github.com/kanishksingh-01/spaceloop	Branch: main (Commit 68d3c46)	Pushed & Up-to-date

10. Verification Matrix & Quality Assurance

All code was validated against an automated test matrix covering units, NLP pipelines, and frontend compilation.

Test Suite	Target Area	Tests Run	Duration	Outcome
test_nlp_listing_assistance_part4.py	Listing Assistance, Word-Boundary Grounding, Multi-Tier Fallback	7 tests	0.069s	100% PASSED
test_listing_assistance.py	Multilingual extraction, factual numbers guardrail, rent preservation	12 tests	0.173s	100% PASSED
test_nlp*.py (All Suites)	Semantic Search, Loop Bot RAG Orchestrator, Entity extraction	31 tests	0.457s	100% PASSED
tsc && vite build	TypeScript typecheck and production rollup bundle compilation	1,550 modules	1.73s	0 ERRORS

11. Git Commit Audit Trail (September 28, 2026)

Commit Hash	Message / Engineering Action
68d3c46	build(dist): deploy updated frontend bundle to public for Vercel production hosting
3cf521d	style(header): refine portal indicator into compact native Host Mode pill
bdb433e	refactor(nav): move List a Space from Host navbar to hamburger menu
f725ea8	feat(nlp): implement part 4 listing assistance with strict zero-hallucination and host review
a2e1a37	style(architecture): fix bottom section layout and spacing below team cards
2a90b68	style(header): refine portal indicator into compact futuristic mode badge
8c83d47	refactor(nav): clean up host portal navigation into hamburger menu
df87a1c	refactor(nav): clean up seeker portal navigation into hamburger menu
32dee79	feat(nlp): implement Part 3 RAG for Loop Bot with grounded LLM synthesis and fallback
4e33f1e	feat(team): replace profile photos for all 4 team members on Architecture page
0db4d85	feat(nlp): implement Part 1 Loop Bot foundation with clean formatting and anti-hallucination

SpaceLoop System Certification
Engineering Sign-off: All tasks specified across SpaceLoop NLP Parts 1–4, Navigation cleanups, and portal design refinements have been implemented, tested, and synced live to production edge servers. The codebase remains compliant with Section 52 Easements Act and strict zero-hallucination policies.